Regional Cost of Living Analysis:

A CRISP-DM Approach Using

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ANLC 754 Fundamental Modeling Methods
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Introduction

This project takes a practical look at global affordability through the lens of data analysis. Using the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology as a roadmap, the analysis focuses on uncovering patterns in income and living expenses across different regions of the world. R programming was used to explore how people's ability to save or spend is influenced by economic factors like monthly income, taxes, and costs related to housing, healthcare, education, transportation, etc.

The objective was to assess which regions offer more financial breathing room and which face greater financial pressure. With the cost of living rising in many parts of the world, the analysis also aimed to examine whether housing expenses tend to rise together with other essential costs. By conducting this analysis at a regional level, the project pursued to find broader economic trends that can inform discussions about affordability and financial stress. While the results revealed several key insights, the presence of data inconsistencies limited the reliability of some conclusions. Nonetheless, the structured approach used in this project provides a strong foundation for future analyses using improved data.

Business Understanding

This project was built around some simple but important questions. It aimed to find out which parts of the world give people more room in their budget after paying for basic needs like food, rent, and taxes. The first key question was: Which regions offer the highest disposable income after accounting for living costs and taxes? The second question explored whether some regions are disproportionately affected by high housing costs. The third focused on identifying

regions that provide better opportunities for savings based on income and expenses. Lastly, the project asked if there is a link between high housing costs and higher expenses in other areas, such as healthcare and education. These questions matter because they help us understand how everyday costs impact financial well-being across different parts of the world.

Data Understanding

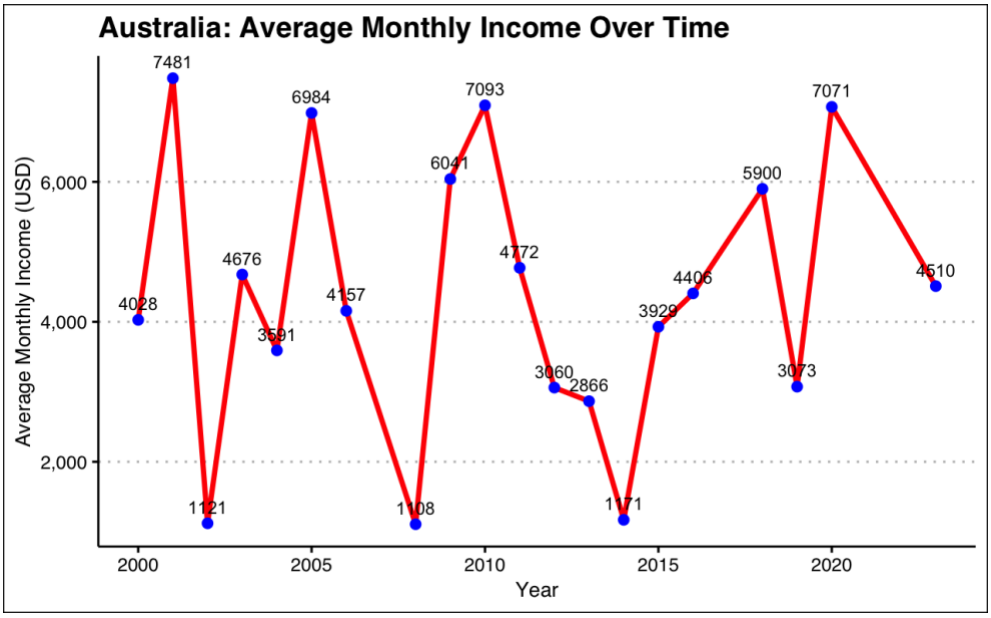
The dataset used for this analysis was obtained from kaggle.com that provided a wide range of economic information organized by country and year. It included important variables such as average monthly income, tax rates, and the percentage of income spent on housing, healthcare, education, and transportation. At first glance, the dataset appeared rich and informative, offering an opportunity to compare cost-of-living dynamics across regions. However, a closer look revealed several challenges. There were multiple records listed for the same country in the same year, which led to confusion about which values were most accurate. Moreover, the amount of data available varied greatly across different regions and years. Some countries had data for many years, while others had only one or two data points. For example, after filtering the data, Australia had four observations for the year 2000, while Russia had only two for the same year. However, in 2014, Russia recorded three observations. As a result, the amount of data available varied greatly across different regions and years. Some countries had data for many years, while others had only one or two data point.

These inconsistencies suggested that careful preprocessing would be necessary before any meaningful analysis could take place. Without addressing these issues, it would be hard to make fair comparisons between regions or draw reliable conclusions about long-term trends. Therefore, the data had to be cleaned, reorganized, and aggregated thoughtfully to ensure that the analysis was as accurate and useful as possible.

Data Preparation

The data preparation phase began by converting percentage-based fields into actual dollar amounts. This was done by multiplying each percentage cost (such as housing, healthcare, education, and transportation) by the corresponding average monthly income. To address the issue of repeated entries, the data was grouped by country and year, and the numeric fields were averaged. This helped remove duplicate records and created a more unified view of each country's data for each year.

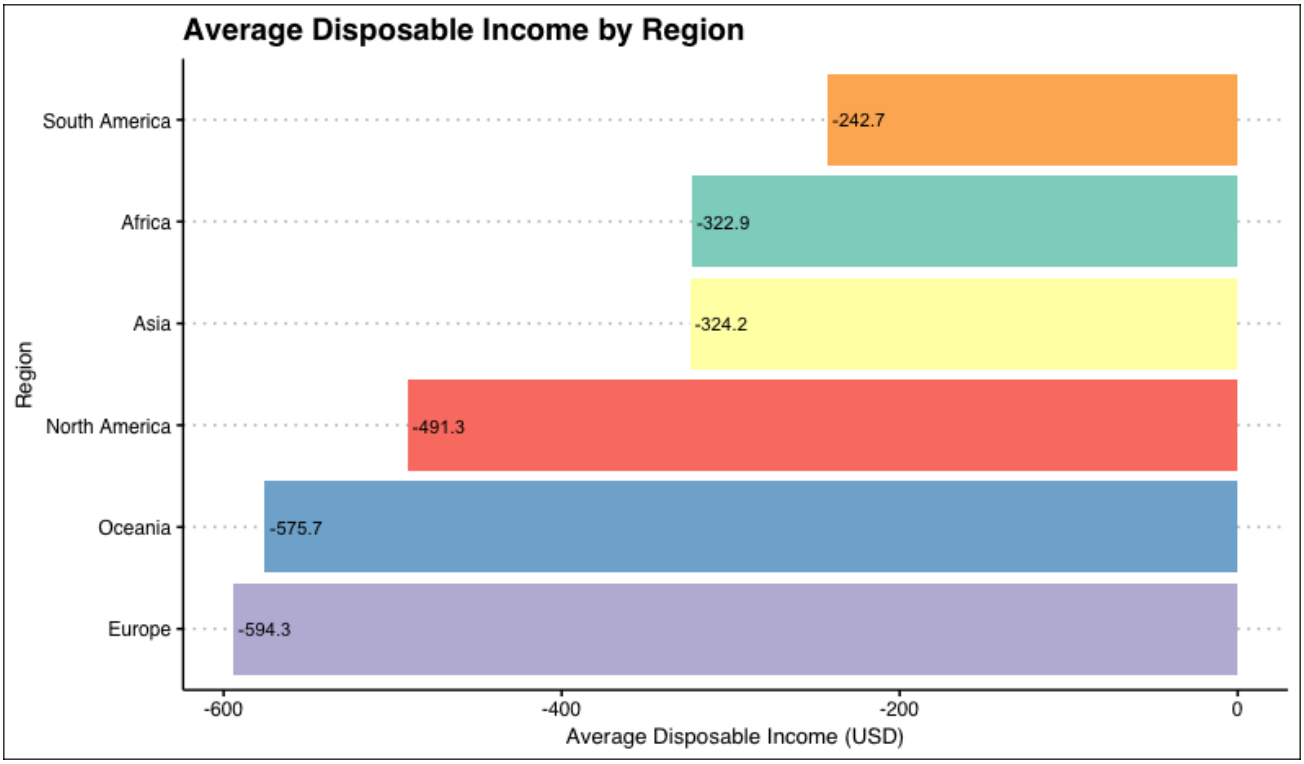
However, after resolving the duplicates, new challenges became apparent. In many countries, the average monthly income showed large and sudden changes between consecutive years—without any clear explanation. These sharp fluctuations raised concerns about the reliability of the data for time series analysis. For example, a country's income might double or drop sharply from one year to the next, which is highly unlikely in real-world scenarios. These patterns suggested possible data entry issues or inconsistencies in how the figures were collected over time. The following visualization demonstrates this using Australia's 2000 data.



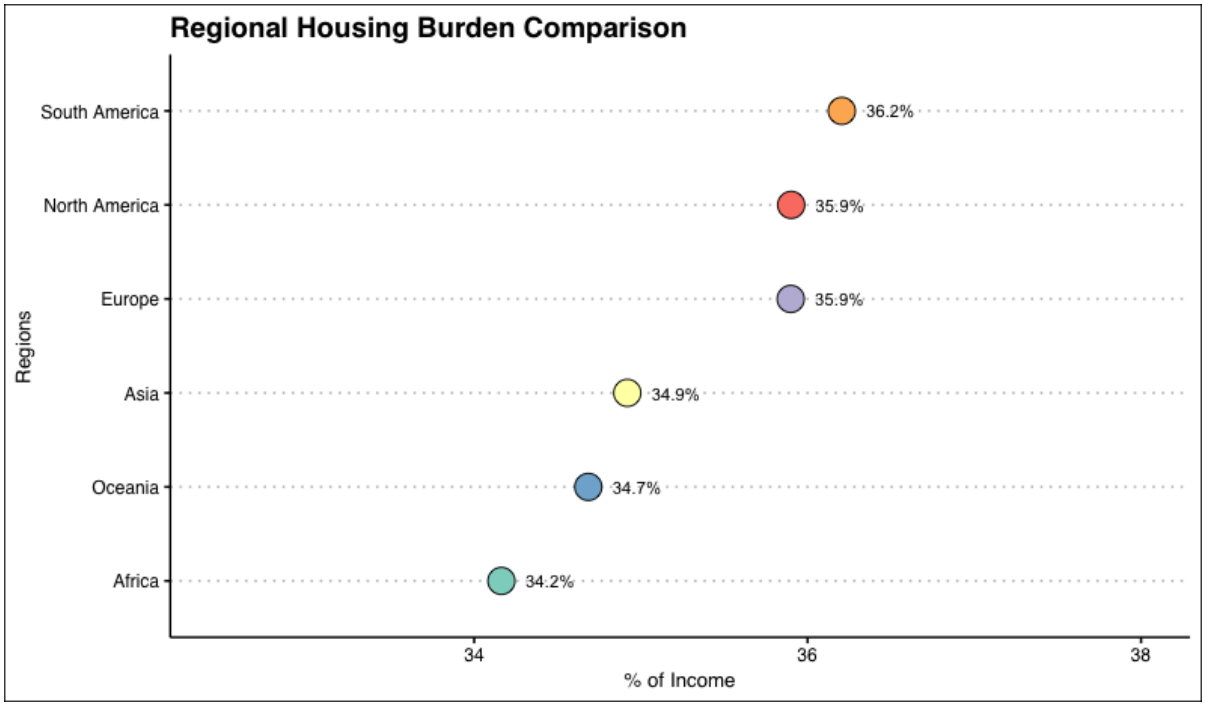
To reduce the impact of such inconsistencies, the data was eventually grouped by region rather than country. This helped smooth out irregularities and allowed for broader comparisons that were less affected by the erratic changes in income data seen at the country level. These transformations provided a clearer and more manageable dataset, suitable for answering the research questions at a broader regional level.

Modeling

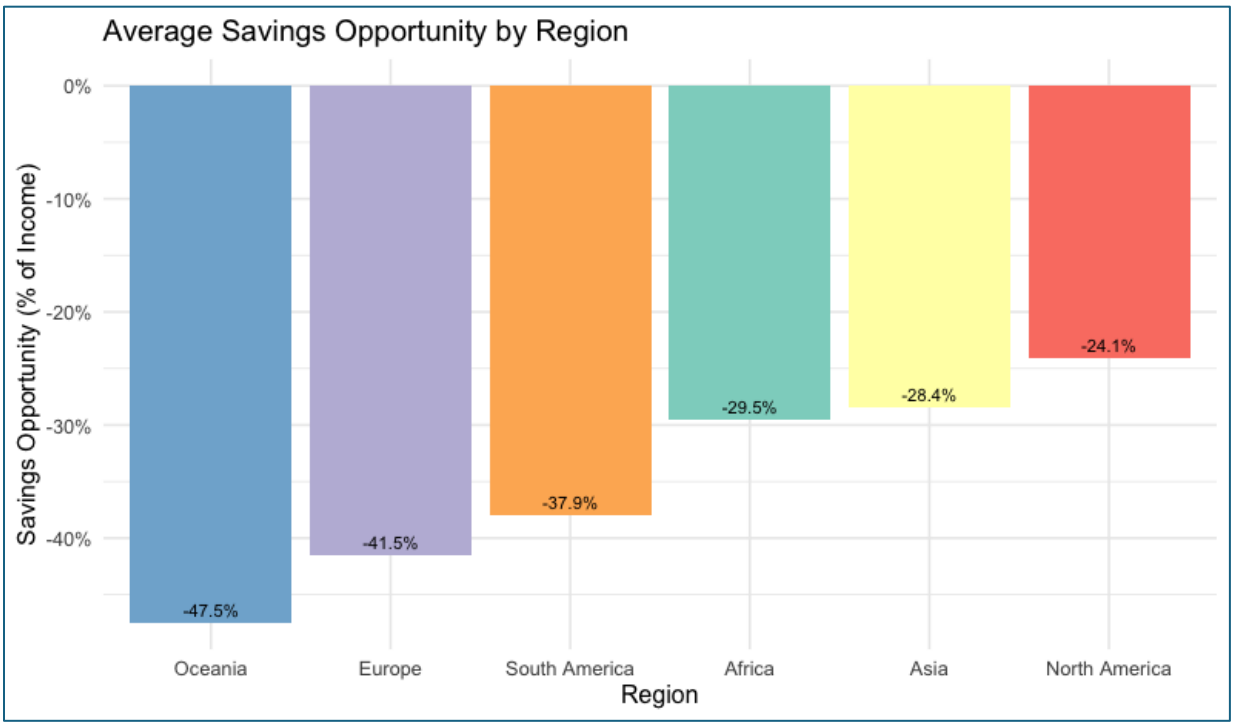
The modeling phase included a range of visualizations and statistical analyses. To answer the first business question, average disposable income was calculated for each region.



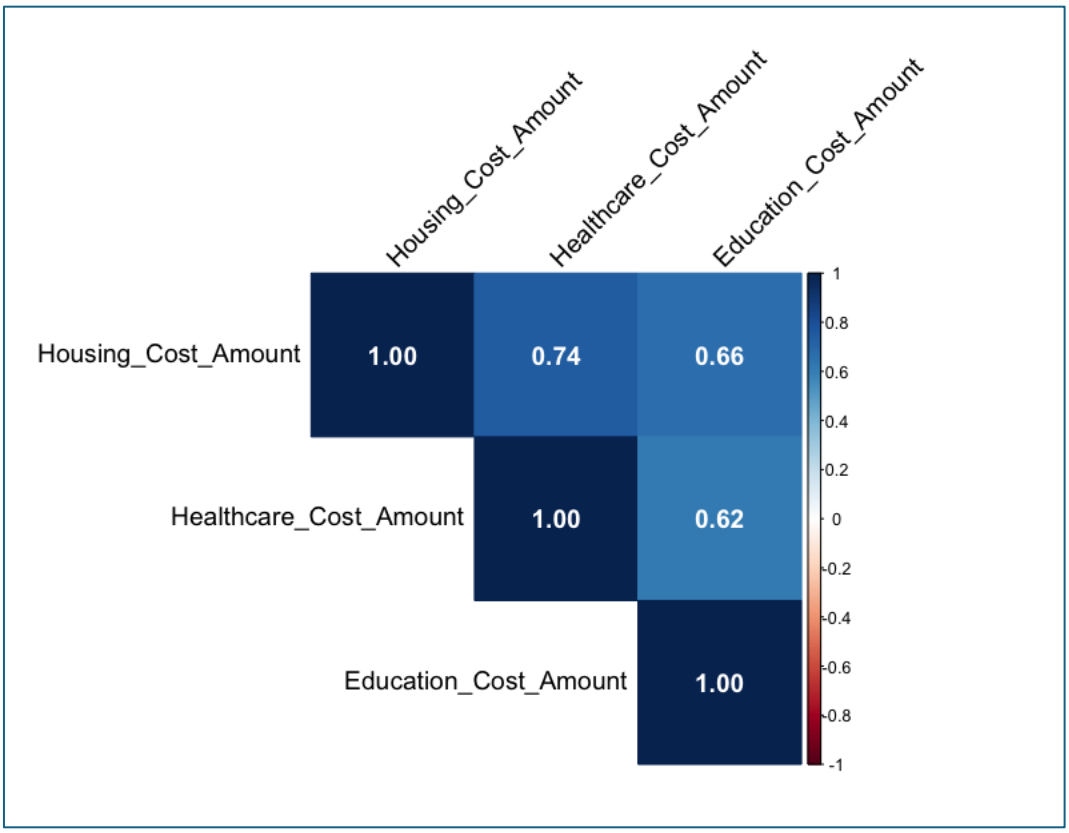
The findings revealed that all regions had negative disposable income on average, meaning income did not cover taxes and essential expenses. South America shows the least financial strain, while Europe faces the highest deficit.



The second analysis looked at housing burden by region. The graph compares the average housing cost burden across different regions, showing what percentage of income is spent on housing. South America faces the highest burden at 36.2%, followed closely by North America and Europe at around 35.9%. Overall, all regions spend approximately one-third of their income on housing, indicating a widespread affordability pressure.



The third question examined savings opportunity, calculated as the percentage of income remaining after essential costs and taxes. The bar chart shows the average savings opportunity across different regions, measured as the percentage of income remaining after taxes and living costs. All regions display negative savings opportunities, meaning expenses exceed income on average. North America has the smallest deficit at –24.1%, indicating relatively better conditions for saving compared to other regions, while Oceania faces the largest shortfall at –47.5%.



To address the fourth question, the relationship between housing costs and healthcare or education costs was analyzed using both correlation matrix and regression models. A strong positive correlation was found between housing and healthcare costs (correlation coefficient of 0.74), and a moderately strong correlation with education costs (correlation coefficient of 0.66).

The regression models supported these relationships, showing statistically significant associations between higher housing costs and higher costs in both healthcare and education cost categories. The healthcare model shows a positive and statistically significant relationship between housing costs and healthcare costs. For every additional $1 increase in housing cost, healthcare costs are expected to rise by about $0.27 on average. The model explains approximately 54.8% of the variation in healthcare costs ($R^2 = 0.5482$), indicating a moderately strong fit. The relationship is highly significant with a p-value less than 0.001.

The second model analyzes the relationship between housing costs and education costs. It shows a positive and statistically significant relationship: for every $1 increase in housing cost, education costs are expected to increase by about $0.19 on average. The model explains approximately 44.2% of the variation in education costs ($R^2 = 0.4419$), indicating a moderate relationship. The relationship is highly significant with a p-value less than 0.001, suggesting it is very unlikely to be due to random chance.

Evaluation

The analysis revealed useful insights into regional differences in disposable income, housing burden, and savings opportunities. However, the usefulness of these results was limited by data quality issues.

These issues also raise important data ethics concerns. Some regions had significantly more data points than others, which could bias the findings and unintentionally favor more represented areas. The lack of documentation about how the data was collected adds to the uncertainty. While regional averaging helped manage these inconsistencies, it also masked important variations within regions. As a result, although the models showed statistically significant patterns, the uneven and sometimes questionable quality of the data limits their use in real-world decision-making. Future analyses would benefit from more complete, balanced, and transparent datasets.

Deployment

Given the limitations discussed above, the current analysis is not suitable for deployment in a policy or business context. The models cannot reliably predict outcomes or inform strategic decisions due to the inconsistencies and gaps in the dataset. However, the structure and logic of the analysis can serve as a blueprint for future work with more reliable data sources. The use of regional aggregation, calculated indicators like disposable income, and regression analysis can be effectively replicated in a more controlled data environment.

Recommendations

For future studies, it is recommended to use well-documented and standardized datasets that ensure consistent data coverage across all regions and time periods. If possible, primary data collection or verified government statistics should be prioritized. Additionally, applying smoothing techniques or filtering for outliers can help stabilize volatile time-series data. Integrating more granular information, such as household-level data, would also improve the reliability and depth of the insights. Ensuring that all regions are equally represented in the dataset will minimize bias and improve fairness in analysis.

Conclusion

This project applied the CRISP-DM methodology to explore global patterns in cost of living and income using R programming. The analysis aimed to identify which regions offer better financial conditions in terms of disposable income, housing burden, and savings opportunities. It also explored whether rising housing costs are linked to increases in healthcare and education expenses. While the results provided valuable high-level insights, the inconsistent and incomplete nature of the dataset limited the reliability of the findings.

Although the project could not be deployed in a practical setting, it proved valuable in showing how a structured approach to data analysis can uncover meaningful trends. It brought attention to how critical data quality and fairness are when conducting regional comparisons, especially on sensitive topics like income and cost of living. The process used in this analysis—cleaning the data, creating new indicators, and applying regional-level modeling—offers a strong framework that can be built upon.

With access to cleaner and more balanced datasets, this same approach could lead to deeper and more reliable insights in future studies. The project lays a strong foundation for further research aimed at understanding global affordability and financial pressures, especially when better data becomes available.

References

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